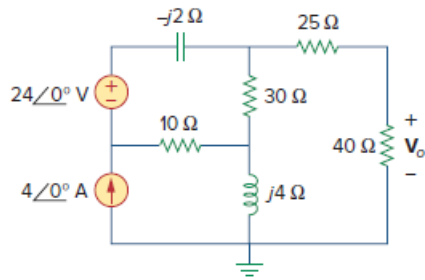


Problem

Use *PSpice* or *MultiSim* to determine V_o in the circuit of Fig. 10.124. Assume $\omega = 1$ rad/s.

Figure 10.124:



Step-by-step solution

Step 1 of 9

Draw the following circuit diagram:

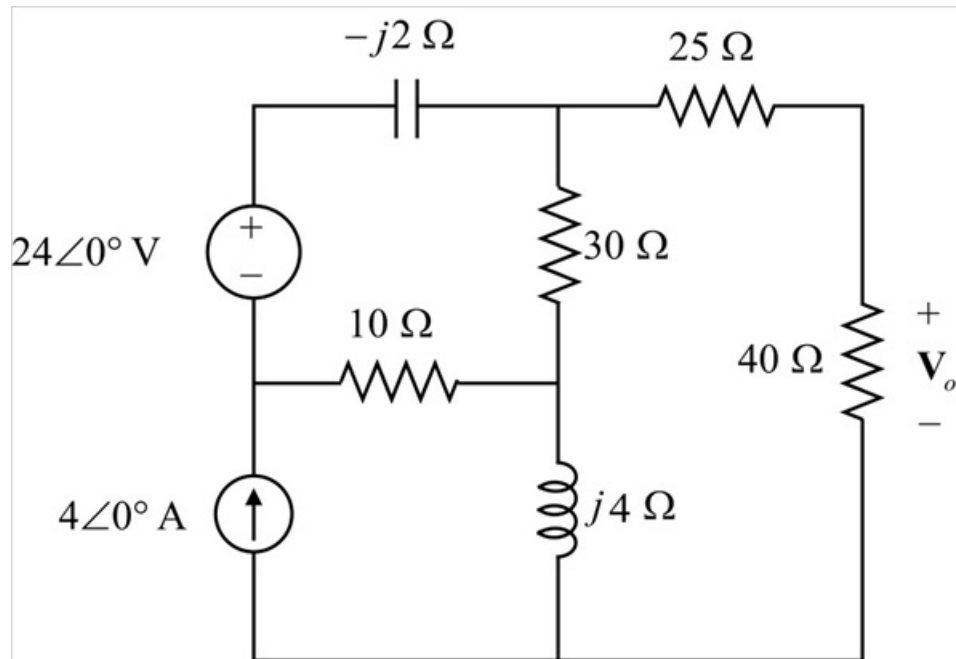


Figure 1

Step 2 of 9

Draw the modified circuit with notations:

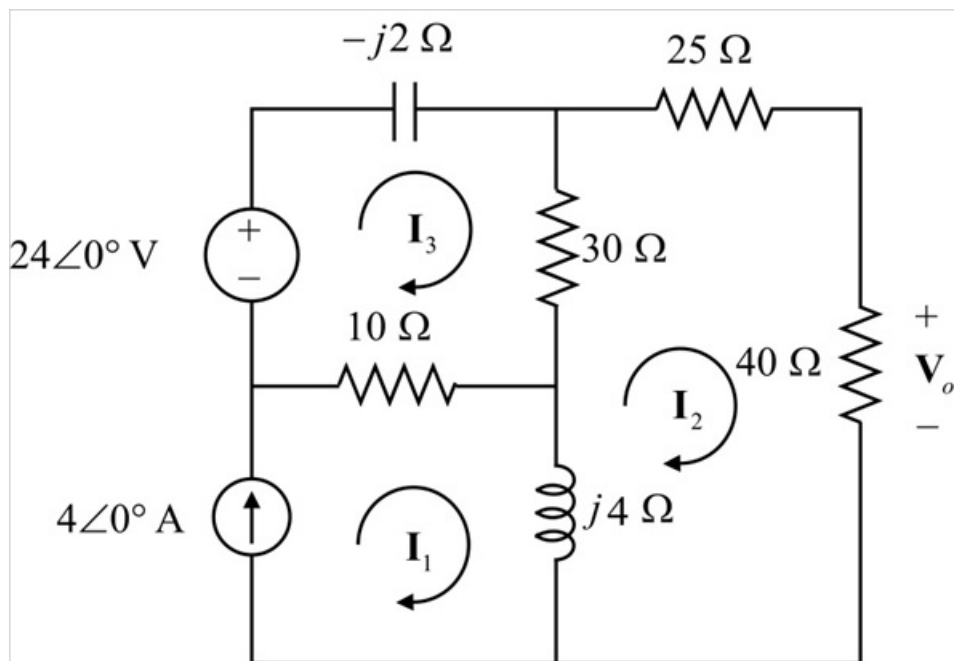


Figure 2

Step 3 of 9

For Mesh 1, the current flowing is $I_1 = 4 \text{ A}$.

Apply Kirchhoff's voltage law for the mesh 3 for the circuit in Figure 2.

$$-24 + I_3(-j2) + (I_3 - I_2)(30) + (I_3 - I_1)10 = 0$$

$$I_3(10 - j2 + 30) = 24 + I_1(10) + I_2(30)$$

$$I_3(40 - j2) = 24 + 10I_1 + 30I_2$$

$$I_3 = \frac{24 + 10I_1 + 30I_2}{(40 - j2)}$$

Substitute 4 A for I_1 in $I_3 = \frac{24 + 10I_1 + 30I_2}{(40 - j2)}$.

$$I_3 = \frac{24 + 10(4) + 30I_2}{(40 - j2)}$$

$$= \frac{64 + 30I_2}{(40 - j2)}$$

Step 4 of 9

Apply Kirchhoff's voltage law for the mesh 2 for the circuit in Figure 2.

$$\mathbf{I}_2(25 + 40) + (\mathbf{I}_2 - \mathbf{I}_1)(j4) + (\mathbf{I}_2 - \mathbf{I}_3)(30) = 0$$

$$\mathbf{I}_2(65 + j4 + 30) = \mathbf{I}_1(j4) + \mathbf{I}_3(30)$$

$$\mathbf{I}_2(95 + j4) = \mathbf{I}_1(j4) + \mathbf{I}_3(30)$$

Substitute 4 A for $\mathbf{I}_1$ and $\frac{64 + 30\mathbf{I}_2}{(40 - j2)}$ for $\mathbf{I}_3$ in $\mathbf{I}_2(95 + j4) = \mathbf{I}_1(j4) + \mathbf{I}_3(30)$.

$$\mathbf{I}_2(95 + j4) = (4)(j4) + \left[\frac{64 + 30\mathbf{I}_2}{(40 - j2)} \right](30)$$

$$\mathbf{I}_2(95 + j4) = (j16) + \frac{1920}{(40 - j2)} + \frac{900\mathbf{I}_2}{(40 - j2)}$$

$$\mathbf{I}_2 \left[(95 + j4) - \frac{900}{(40 - j2)} \right] = (j16) + \frac{1920}{(40 - j2)}$$

$$\mathbf{I}_2 \left[\frac{3808 - j30 - 900}{(40 - j2)} \right] = \frac{32 + j640 + 1920}{(40 - j2)}$$

$$\begin{aligned} \mathbf{I}_2 &= \frac{1952 + j640}{2908 - j30} \\ &= 0.669 + j0.227 \end{aligned}$$

Step 5 of 9

Calculate the output voltage $\mathbf{V}_o$.

$$\mathbf{V}_o = \mathbf{I}_2(40)$$

Substitute $0.669 + j0.227$ for $\mathbf{I}_2$ in $\mathbf{V}_o = \mathbf{I}_2(40)$.

$$\begin{aligned} \mathbf{V}_o &= (0.669 + j0.227)(40) \\ &= 26.756 + j9.08 \\ &= 28.25 \angle 18.73^\circ \text{ V} \end{aligned}$$

Therefore, the output voltage $\mathbf{V}_o$ is $28.25 \angle 18.73^\circ \text{ V}$.

Consider the following data:

The value of ω is 1 rad/s

The value of X_C is $-j2 \Omega$

The value of X_L is $j4 \Omega$

Calculate the value of the capacitor.

$$X_C = -j2$$

$$\frac{-j}{\omega C} = -j2$$

$$C = \frac{1}{2\omega} \quad (\text{since } \omega = 1 \text{ rad/s})$$

$$C = 0.5 \text{ F}$$

Step 6 of 9

Calculate the value of the inductor.

$$X_L = j4$$

$$j\omega L = j4$$

$$L = \frac{4}{\omega} \quad (\text{since } \omega = 1 \text{ rad/s})$$

$$L = 4 \text{ H}$$

Draw the modified circuit diagram:

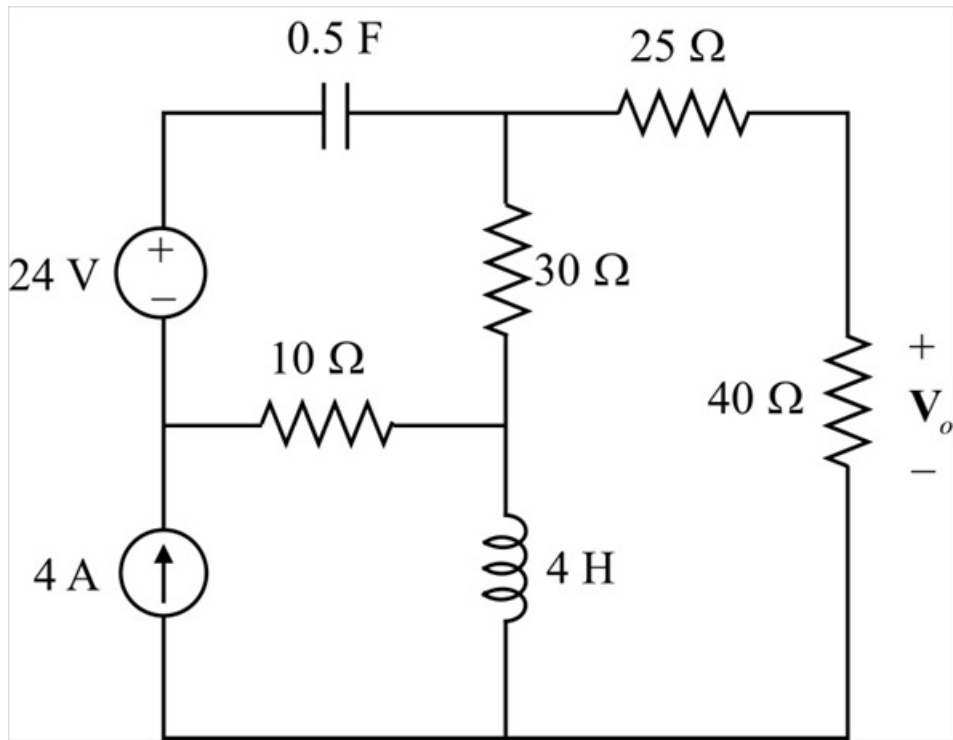


Figure 3

Draw the schematic diagram:

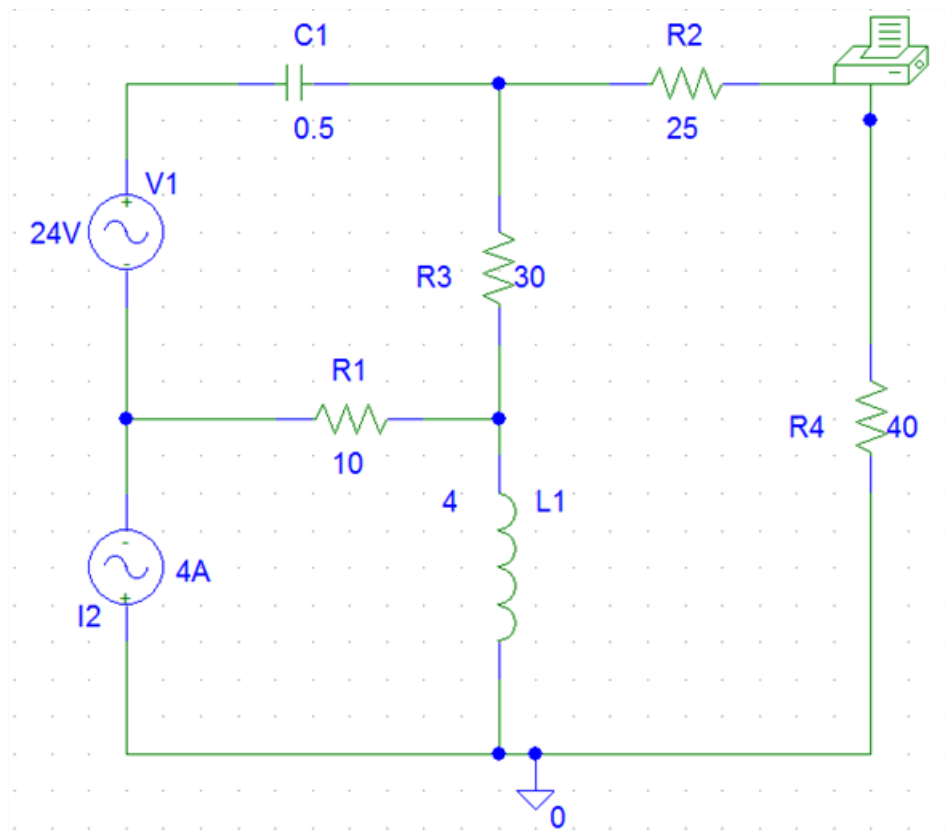


Figure 4

The output after simulating the circuit is as follows:

FREQ VM(\$N_0004) VP(\$N_0004)

1.590E-01 2.825E+01 1.873E+01

Therefore, the output voltage is $28.25 \angle 18.73^\circ \text{ V}$.